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A new characterization of arithmetical categories and Pixley's theorem

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We characterize the arithmetical categories [1] - the exact Mal'tsev categories whose lattices of congruences are distributive - by a simple condition on some suitable pushouts and a finite limit: a regular category is exact arithmetical if and only if, for every triple of regular epimorphisms with common domain, the pairwise pushouts exist and the comparison morphism to the limit of the diagram they form is a regular epimorphism. For a regular category, the validity of this condition for pairs and for triples is equivalent to its validity for n-tuples for all $n \geq 2$, and characterizes arithmetical categories among regular ones. For pairs alone, that is when $n = 2$, the condition reduces to the characterization of exact Mal'tsev categories among regular ones, due to Carboni, Kelly and Pedicchio [2]. For n-tuples with $n \geq 4$ nothing further is obtained: arithmeticity is the third and last rung of that ladder. The proof rests on a categorical form of the Chinese Remainder Theorem; in the exact Mal'tsev context this form of the theorem lies close to results of Hoefnagel on majority categories [3]. As a direct application, we give a new proof of Pixley's characterization of arithmetical varieties [4] using a suitable diagram of free algebras. The main results presented in this talk are proved in the preprint [5].

Keywords: Arithmetical category; Pixley's theorem; the Chinese Remainder Theorem
Subject area: Category theory and universal algebra
MSC2020: 18E13, 08B10, 18E08

References

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[4] A. F. Pixley, *Distributivity and permutability of congruence relations in equational classes of algebras*, Proceedings of the American Mathematical Society **14** 105–109, 1963.

[5] M. Gran, *A new characterization of arithmetical categories and Pixley's theorem*, arXiv:2609.06248, preprint, 2026.